

Generative AI and Self-Regulated Learning in Education: A Scientometric Review

Generative AI and Self-Regulated Learning in Education

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Abstract

This study presents a scientometric review of the intersection between Generative Artificial Intelligence (Generative AI) and Self-Regulated Learning (SRL) in educational contexts. Based on 79 articles retrieved from Web of Science Core Collection and analyzed using VOSviewer. The keyword co-occurrence analysis revealed five thematic clusters: 1) SRL theory and cognition (metacognition, cognitive load), 2) AI design and learner experience (learning companion, human-AI interaction), 3) higher education practices (feedback, academic writing), 4) instructional design with LLMs (prompt engineering, K-12), and 5) methodological and regional studies (critical thinking, AI in China). This review contributes to the theoretical integration of SRL and AI-enhanced learning. The study identifies three strategic directions: 1) embedding metacognitive scaffolds in AI interfaces rather than providing direct answers, 2) implementing transparency mechanisms to develop critical AI literacy, and 3) evolving assessment practices to evaluate learners' AI regulation capabilities.

CCS Concepts

• **Human-centered;** • **computing Human computer interaction (HCI);** • **Interaction techniques;** • **Text input;**

Keywords

Generative AI, Self-Regulated Learning, Educational Technology, Bibliometric Analysis, ChatGPT, Scientometrics

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1 Introduction

In recent years, generative artificial intelligence (Generative AI), particularly large language models (LLMs) such as ChatGPT and GPT-4, has become a transformative force across various domains of knowledge, including education. These AI technologies have sparked significant discourse regarding their pedagogical utility and ethical implications in contemporary learning environments, fundamentally challenging traditional approaches to teaching and learning[1].

The rise of generative AI coincides with a renewed focus on self-regulated learning (SRL), a theoretical framework that emphasizes the learner's active control over cognitive, motivational, and behavioral aspects of learning[2, 3]. With the increasing integration of AI-powered systems in education since 2022, a critical research frontier has emerged: how can generative AI be designed and used to scaffold, enhance, or potentially hinder learners' self-regulatory capabilities?

Despite the growing interest in both domains, several research gaps persist. First, most studies examine generative AI and SRL separately, focusing either on AI's technical affordances[4, 5] or on SRL in conventional learning settings[6]. Second, few empirical studies analyze how learners regulate cognition and emotion when using generative AI tools. Third, there is a lack of systematic mapping to identify research hotspots, theoretical foundations, and emerging trajectories in this interdisciplinary field. Recent comprehensive reviews have traced AI's evolution across multiple domains, providing essential context for understanding current educational applications. [7] examined AI's trajectory from narrow task-specific systems to contemporary generative models, highlighting persistent challenges in explainability, bias, and human-AI collaboration. Their analysis underscores that educational AI applications inherit these broader technical and ethical tensions, making the intersection with self-regulated learning particularly complex. Unlike earlier AI systems that operated as static knowledge repositories, generative AI's dynamic, context-sensitive outputs require learners to develop new regulatory strategies for critical evaluation and appropriate reliance.

To answer these questions, we conduct a bibliometric mapping of research on generative AI and SRL in education. This approach allows us to visualize knowledge structures, trace thematic clusters, and reveal emerging trajectories with greater objectivity and reproducibility than narrative reviews alone[8].

The findings from this study serve three primary objectives.

- Providing a comprehensive overview of the current state and evolution of research on generative AI and SRL in educational contexts;
- Identifying key research clusters, influential contributors, and theoretical foundations through systematic network analysis;
- To provide strategic guidance for researchers, instructional designers, and policymakers seeking to leverage generative AI for self-regulated learning across formal and informal educational settings.

2 Methods

2.1 Research Design

This study uses bibliometric analysis to map research at the intersection of generative AI, SRL, and education. Bibliometric analysis is a quantitative method for examining publication metadata (e.g., titles, abstracts, keywords, citations) to uncover knowledge structures, trends, and influential contributors in a given field[8, 9].

To facilitate data visualization and network mapping, we employed VOSviewer (version 1.6.19), a widely used software tool for constructing and interpreting bibliometric networks such as keyword co-occurrence, author co-citation, and bibliographic coupling[10]. VOSviewer was selected for its robust clustering algorithms and intuitive visualization capabilities. The specific parameters used in this study included: 1) VOS clustering technique with a resolution parameter of 1.0; 2) LinLog/modularity normalization for network layout; 3) association strength as the similarity measure for keyword co-occurrence; and 4) fractional counting method for handling multi-authored publications. These parameter settings were chosen based on established best practices in bibliometric research to ensure optimal cluster identification and visual clarity.

2.2 Data Source and Collection

Data were retrieved exclusively from the Web of Science Core Collection (WoS), which provides structured bibliographic records for peer-reviewed journals with comprehensive citation tracking capabilities. The decision to use WoS as the sole database was deliberate and methodologically justified. First, WoS maintains the most rigorous quality control standards in academic indexing, ensuring that only high-impact, peer-reviewed publications are included. Second, WoS provides the most complete and accurate citation linkage data, which is essential for co-citation and bibliographic coupling analyses. Third, using a single database eliminates technical challenges associated with merging records from multiple sources, such as duplicate detection, inconsistent metadata formats, and incompatible citation indexing systems. This approach aligns with established practices in bibliometric research, where single-database strategies are preferred when analytical rigor and citation network accuracy are prioritized[9].

To identify relevant literature, we developed a comprehensive Boolean search strategy that combines three conceptual domains: 1) artificial intelligence and machine learning technologies, 2) self-regulated learning and metacognitive processes, and 3) educational contexts. The final search query was constructed as follows: Search Strategy: TS=(“artificial intelligence” OR “machine learning” OR

“natural language processing” OR “intelligent tutoring” OR “adaptive learning” OR “educational AI”) AND (“self-regulated learning” OR “self-regulation” OR “metacognition” OR “learning strategies” OR “self-directed learning”) AND (“education” OR “learning” OR “teaching” OR “pedagogy” OR “instructional design”). The time-span was set from January 2010 to September 2025 to capture the evolution of research in AI-enhanced education.

2.3 Data Preprocessing and Inclusion Criteria

Before visualization, preprocessing included: 1) removing duplicates and incomplete conference abstracts; 2) limiting to English-language publications; 3) retaining only articles and reviews while excluding non-scholarly items; and 4) setting a minimum keyword occurrence threshold of five for mapping.

The threshold of five occurrences was selected based on multiple methodological considerations. First, this threshold represents a standard practice in bibliometric research using VOSviewer, as recommended in the software’s technical documentation and widely adopted in published scientometric studies. Second, given our corpus of 79 articles, a threshold of five ensures that only keywords with sufficient representativeness are included in the network, thereby reducing noise while maintaining meaningful thematic patterns. Third, we conducted sensitivity analyses by testing thresholds of 3, 5, and 7 occurrences. Results showed that a threshold of 3 produced an overly dense network with limited interpretability, while a threshold of 7 resulted in the exclusion of several theoretically important terms. The threshold of five achieved an optimal balance between network comprehensiveness and visual clarity, yielding 45 keywords organized into five interpretable thematic clusters. This methodological decision enhances both the validity of cluster formation and the practical utility of the visualizations.

After applying these criteria and manual screening for relevance, 79 articles were retained for analysis. Before visualization, additional preprocessing included: 1) standardizing author keywords and removing duplicates; 2) merging synonymous terms (e.g., “AI” and “artificial intelligence”); 3) applying the aforementioned minimum keyword occurrence threshold of 5 for network mapping; and 4) verifying citation linkages for accuracy. These stringent inclusion criteria and preprocessing steps were designed to ensure data quality and analytical rigor, consistent with established practices in bibliometric research[11, 12].

3 Results and Discussion

This section presents a comprehensive scientometric analysis of 79 articles examining the intersection of Generative AI and Self-Regulated Learning in education. Of the 79 articles analyzed, 54 (68.4%) were published between 2022-2025, 18 (22.8%) between 2018-2021, and 7 (8.8%) between 2010-2017. Six types of bibliometric visualizations were generated using VOSviewer, each providing distinct insights into the knowledge domain structure, research trends, and scholarly influence patterns.

3.1 Keyword Co-occurrence Mapping

Keyword co-occurrence mapping produced five clusters (Figure 1). Cluster 1 relates to SRL theory and cognition (e.g., metacognition,

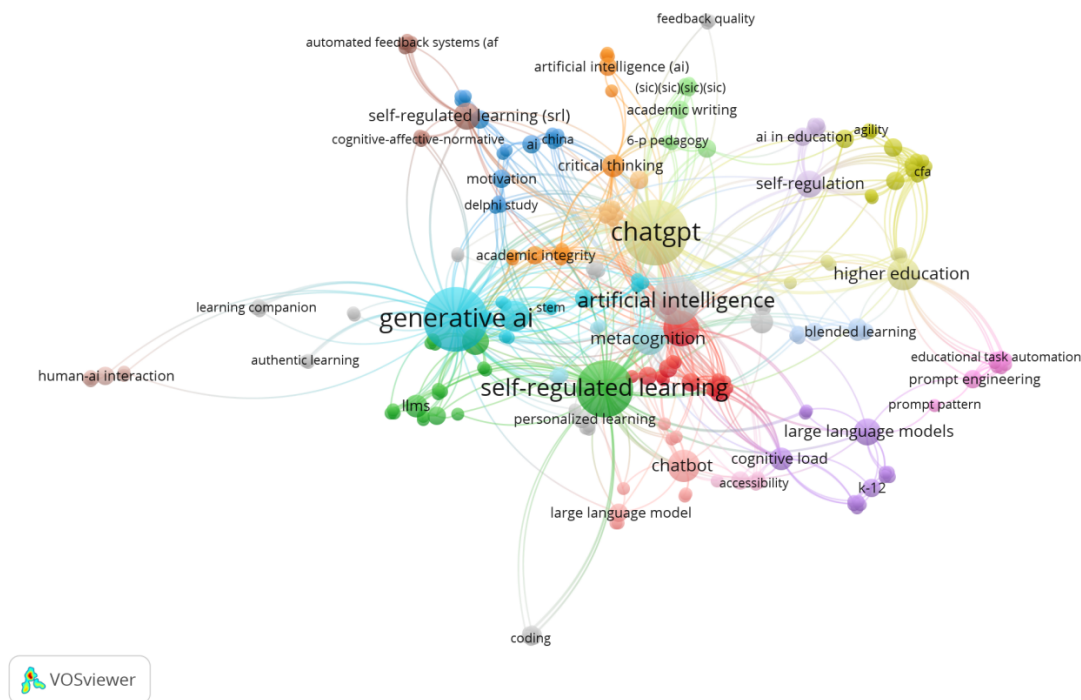


Figure 1: Author Keyword Co-occurrence Network

cognitive load); Cluster 2 to AI design and learner experience (e.g., learning companion, human-AI interaction); Cluster 3 to higher education practices (e.g., feedback, academic writing); Cluster 4 to instructional design with LLMs (e.g., prompt engineering, automation, K-12); and Cluster 5 to methodological and regional studies (e.g., critical thinking, Delphi, AI in China). Overall, the field is evolving from theoretical frameworks toward practical applications, methodological innovation, and ethical considerations.

The clustering patterns observed in our analysis align with broader AI development trajectories documented by [7], who noted AI's shift from symbolic reasoning systems to data-driven, probabilistic models. This evolution is particularly evident in Cluster 4 (instructional design with LLMs), where prompt engineering represents a fundamentally new mode of human-AI interaction compared to earlier rule-based tutoring systems.

3.2 Bibliographic Coupling by Source

Bibliographic coupling analysis identified 23 journals with significant interconnections (Figure 2). The coupling shows that *British Journal of Educational Technology*, *Computers & Education*, and *Education Sciences* are the most closely related and frequently used journals in this field. Other high-influence sources included *Computers in Human Behavior*, *Journal of Educational Computing Research*, and *Plos One*. These journals are widely known for publishing studies related to AI in education and digital learning.

The clustering indicates that most of the literature is concentrated in educational technology and interdisciplinary journals, suggesting that research on generative AI and SRL is situated at the intersection of education, psychology, and computer science.

3.3 Citation Analysis by Country

Citation analysis revealed contributions from 34 countries, with notable concentration in specific regions (Figure 3). The analysis is based on institutional affiliations of authors, with total citations ranging from 5 to 347 per country. As shown in Figure 3, the most dominant countries were China, the United States and Germany. These countries had both high citation counts and collaborative linkages. China and the USA were the most central, indicating that scholars in these countries have contributed significantly to foundational and high-impact research.

Additionally, countries such as Malaysia, Indonesia, and Saudi Arabia appeared in the periphery of the network, indicating emerging interest and research development in these regions. The findings reveal a bipolar research landscape with dominant contributions from China and the USA, supplemented by strategic contributions from European and other Asian countries. This geographic distribution can be attributed to several interconnected factors. First, both China and the USA possess substantial research funding infrastructure and national AI development strategies that prioritize educational application. Second, these countries host leading

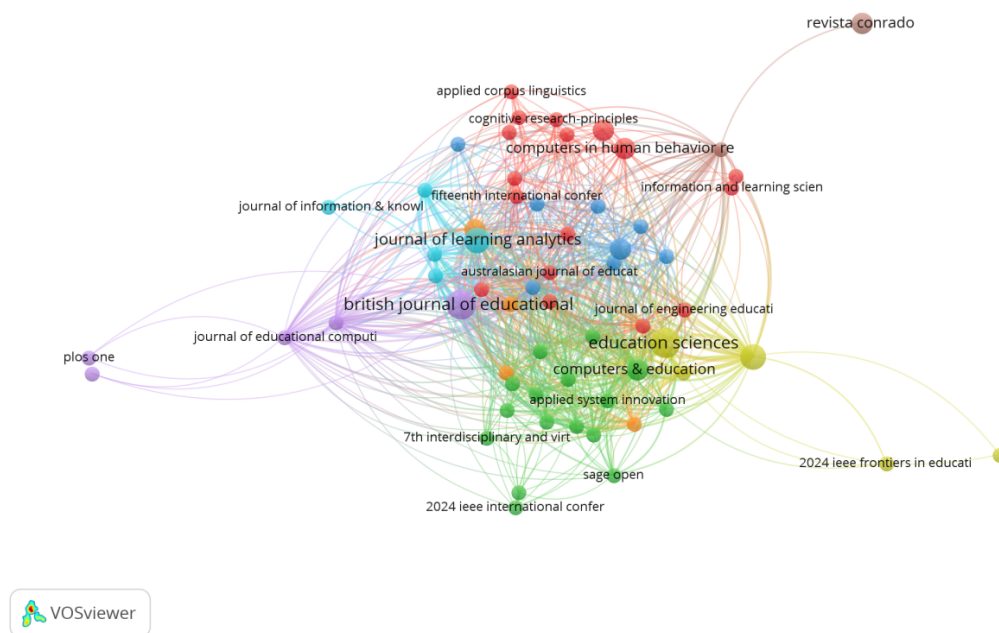


Figure 2: Bibliographic Coupling of Sources

universities with established research centers in educational technology and learning sciences, facilitating high-impact publications. Third, the strong presence of China and Germany reflects their long-standing traditions in rigorous educational research and advanced technological ecosystems. The peripheral positioning of Southeast Asian and Middle Eastern countries suggests emerging research capacity, possibly driven by recent governmental investments in digital education transformation. This distribution reflects not only technological capabilities but also institutional maturity, research funding availability, and strategic prioritization of AI in education within national innovation agendas.

3.4 Citation Analysis of Highly Cited Documents

Analysis of document-level citations identified 25 highly cited articles forming distinct research streams (Figure 4). The temporal distribution shows a marked increase in high-impact publications after 2022. Works by authors such as Fan, Chang, Wu, and Chiu appeared in central positions and were among the most influential.

These studies mostly focused on generative AI applications in higher education, cognitive scaffolding, AI feedback systems, and ethical implications of LLMs in academic writing. This result supports the conclusion that ChatGPT and related technologies have had a strong influence on academic discourse since 2022.

3.5 Co-Citation of Authors

Author co-citation analysis was conducted to identify the most influential scholars in the field. As shown in Figure 5, the most frequently co-cited authors were Zimmerman, Pintrich, Panadero, Bandura, and Winne. These names are well-known for their contributions to SRL theory, cognitive psychology, and social learning.

The presence of authors like Hwang and Chang suggests the emergence of Asian contributions in AI-supported learning design. This analysis confirms that research in this area is grounded in both classical and contemporary psychological theories.

3.6 Co-Citation of References

Figure 6 displays the co-citation network of specific references. Foundational works such as [2, 3] appeared at the center of the network. These articles form the theoretical backbone of SRL and metacognitive research. Recently published works, such as [13] and [14], were also frequently co-cited, reflecting their emerging influence in the study of generative AI and educational innovation. These findings indicate that the field relies heavily on a combination of classic educational psychology literature and newly emerging empirical research on AI applications.

4 Conclusion

This bibliometric analysis of 79 publications (2010-2025) revealed five thematic clusters in generative AI and SRL research: theoretical

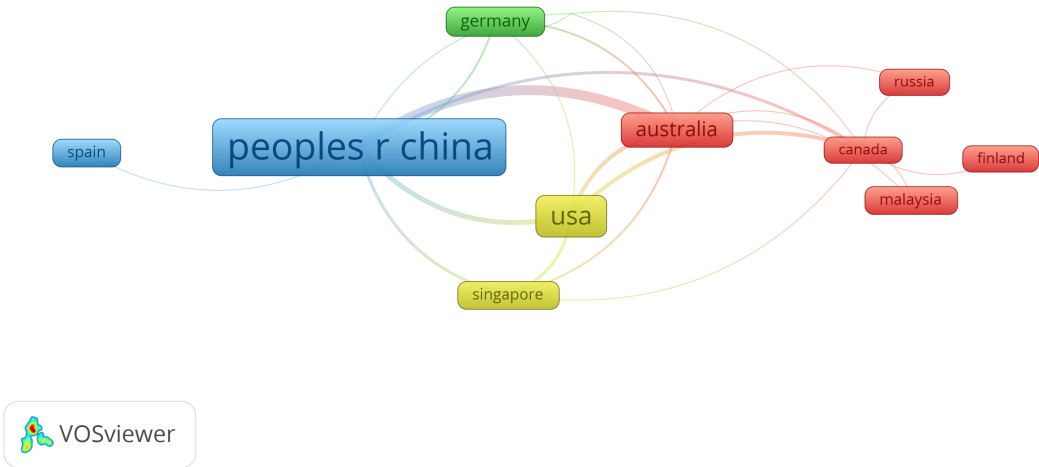


Figure 3: Citation Analysis by Countries

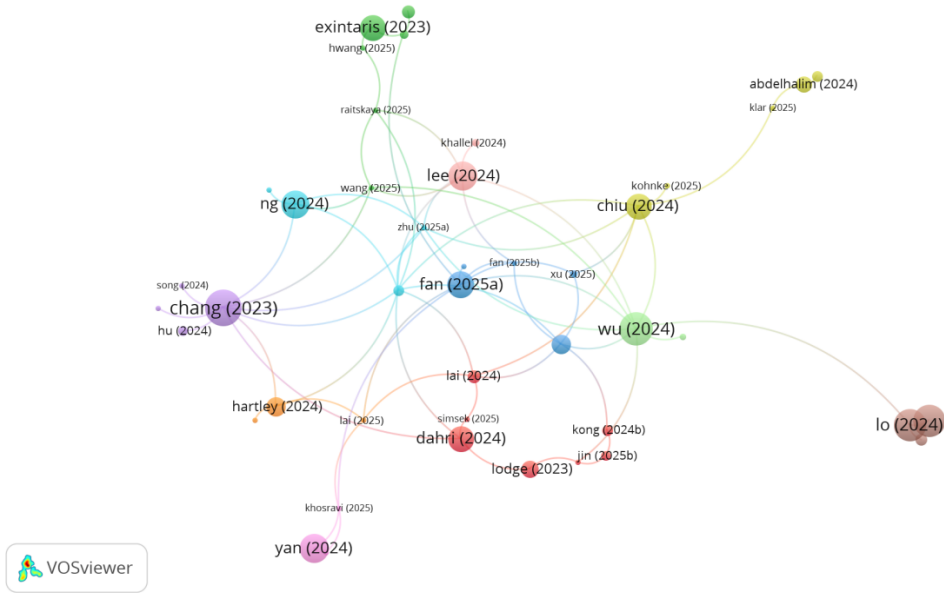


Figure 4: Highly Cited Documents Network

foundations, AI-human interaction design, higher education applications, instructional scaffolding, and methodological innovations. China and the United States lead contributions, with publications surging post-2022 following ChatGPT’s emergence. Foundational works by [2, 3] provide theoretical anchoring, while recent studies

by [13] and [14] shape contemporary discourse on AI-enhanced learning.

Three strategic implications emerge for educational practice. First, AI systems must scaffold rather than replace metacognitive processes through embedded reflection prompts and strategy articulation requirements. Second, ethical frameworks must address

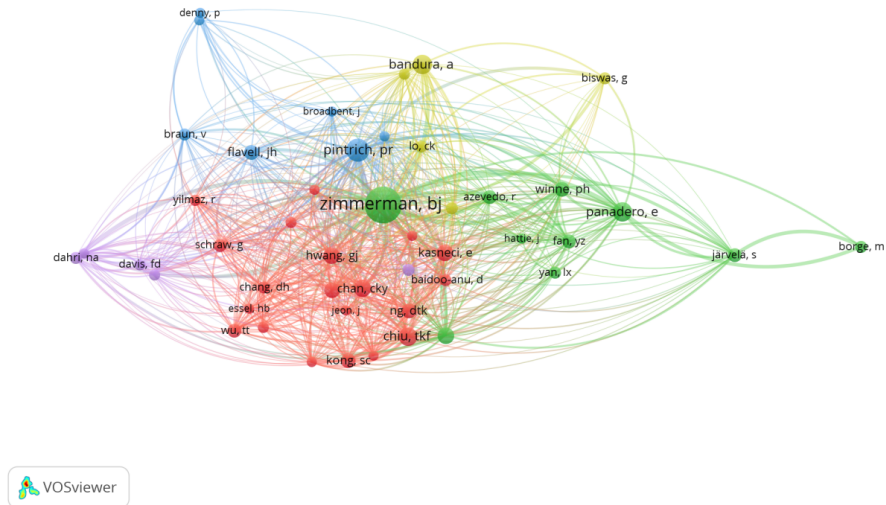


Figure 5: Co-citation Network of Cited Authors

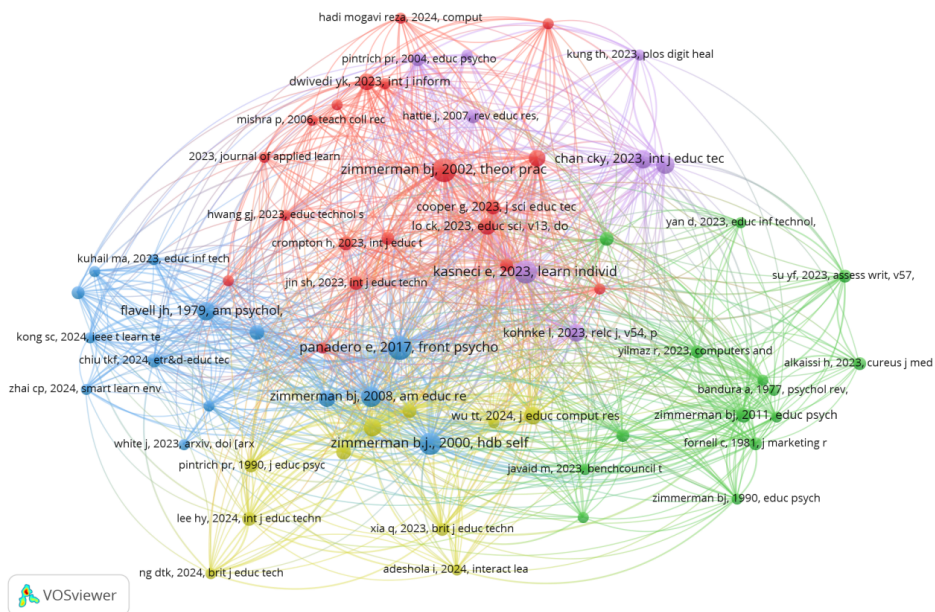


Figure 6: Co-citation Network of Cited References

academic integrity, algorithmic bias, and equitable access through transparency mechanisms and critical AI literacy development. Third, assessment must evolve to evaluate AI regulation capabilities when seeking assistance, how to verify outputs, and how to

integrate feedback iteratively. These implications align with SRL theoretical principles[2] while addressing practical concerns raised by rapid ChatGPT adoption.

Future research should address three gaps identified in this review. First, longitudinal studies examining how sustained AI use impacts learners' self-regulatory development are needed. Second, cross-cultural investigations beyond dominant China-USA contributions would illuminate diverse pedagogical applications and ethical considerations. Third, empirical studies testing specific design principles (e.g., metacognitive prompting vs. direct answer provision) would strengthen evidence-based AI tool development. As generative AI capabilities expand, interdisciplinary collaboration between learning scientists, AI developers, and ethicists becomes essential to ensure these technologies cultivate autonomous, reflective learners rather than passive consumers of algorithmic outputs.

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